



SENSE

Observe state or
context

PROPOSE

Generate candidate
actions or references

ARBITRATE

Approve, modify, reject
or defer

EXECUTE

Send authorized
command

Flow legend

Information flow
(State/context)

Proposal flow
(Candidate actions /
references)

Arbitration flow
(Approve / modify /
reject / defer)

Execution flow
(Authorized command)

1

Human Operator

Situational awareness, high-level intent

2

Planner

Task and trajectory planning

3

Learning Policy

State-to-action mapping (skill or DRL policy)

4

Safety Supervisor

Constraint checking, risk veto or override

5

Low-Level Controller

Tracking and stabilization

6

Actuator Interface

Physical execution

Holds execution authority?

☐ No

☐ No

☐ No

☐ No

Proposal /
arbitration layers

☒ Yes

☒ Yes

Execution-authority
layers

Effective Authority Boundary

*Only layers below this boundary
can commit motion*